

Research note

September 28, 2024

Abstract

This is a research note for implementing quasi-Newton method on minimax problem.

1 Introduction

We focus on the minimax problem:

$$\min_{x \in \mathbb{R}^n} \max_{y \in \mathbb{R}^m} f(x, y) = \min_{x \in \mathbb{R}^n} \Phi(x) \quad (1)$$

where $f : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ is twice continuously differentiable, i.e. $f \in \mathcal{C}^2$. We define the local optimal solution of this problem as follow:

Definition 1 (strict local minimax(SLmM), Evtushenko (1974a)). *(x^*, y^*) is a strict local minimax point (SLmM) of a twice differentiable function f if it is a stationary point and $\partial_{yy}f(x^*, y^*) < 0$ and $D_{xx}f(x^*, y^*) > 0$*

2 Preliminary

We focus on the minimax problem:

$$\min_{x \in \mathbb{R}^n} \max_{y \in \mathbb{R}^m} f(x, y) = \min_{x \in \mathbb{R}^n} \Phi(x) \quad (2)$$

where $\Phi(x) = \max_{y \in \mathbb{R}^m} f(x, y)$. Suppose we can find a solution of the maximization problem $y(x) = \arg \max_{y \in \mathbb{R}^m} f(x, y)$, we have $\Phi(x) = f(x, y(x))$, and we can see the minimax problem as a minimization problem $\min_{x \in \mathbb{R}^n} \Phi(x)$ and a maximization problem $\max_{y \in \mathbb{R}^m} f(x, y)$.

For the maximization problem, we have many ways to find the approximate solution such as gradient ascent, Newton method and quasi-Newton method.

For the minimization problem, we need to analyze the function $\Phi(x) = f(x, y(x))$ furthermore. We know that

$$\nabla \Phi(x) = \nabla_x f(x, y(x)) + \nabla y(x) \nabla_y f(x, y(x)) = \nabla_x f(x, y(x)) \quad (3)$$

since $y(x)$ solve $\max_{y \in \mathbb{R}^m} f(x, y)$ so $\nabla_y f(x, y(x)) = 0$. Therefore, for the gradient descent step, the update will be $x^{k+1} = x^k - \alpha_k \nabla \Phi(x^k) = x^k - \alpha_k \nabla_x f(x^k, y(x^k))$. If we want to take a Newton update, we need to calculate the Hessian $\nabla^2 \Phi(x)$. From $\nabla \Phi(x) = \nabla_x f(x, y(x))$, we know that

$$\begin{aligned} \nabla^2 \Phi(x) &= \nabla_{xx} f(x, y(x)) + \nabla y(x) \nabla_{yx} f(x, y(x)) \\ &= \nabla_{xx} f(x, y(x)) - \nabla_{xy} f(x, y(x)) (\nabla_{yy} f(x, y(x)))^{-1} \nabla_{yx} f(x, y(x)) \end{aligned} \quad (4)$$

The second equality use the fact that

$$\begin{aligned} \nabla_{xy} f(x, y(x)) + \nabla y(x) \nabla_{yy} f(x, y(x)) &= 0 \\ \nabla y(x) &= -\nabla_{xy} f(x, y(x)) (\nabla_{yy} f(x, y(x)))^{-1} \end{aligned}$$

since $\nabla_y f(x, y(x)) = 0$. By using [equation 3](#), the Newton update will be:

$$x^{k+1} = x^k - (\nabla^2 \Phi(x^k))^{-1} \nabla \Phi(x^k) \quad (5)$$

Since the Hessian used in Newton method might be time consuming, we can try to use quasi-Newton method to approximate the Hessian we need:

$$x^{k+1} = x^k - H(x^k)^{-1} \nabla \Phi(x^k) = x^k - B(x^k) \nabla \Phi(x^k) \quad (6)$$

where $H(x)$ is the approximate Hessian and $B(x)$ is the inverse of approximate Hessian. Here, we have many option for the approximation such as BFGS.

3 BFGS

In BFGS, we approximate the Hessian by the step difference and the gradient difference. Normal BFGS update for minimization problem $\min_{x \in \mathbb{R}^n} f(x)$ is

$$x_{k+1} = x_k - H_k \nabla f_k \quad (7)$$

where H is the inverse Hessian approximation:

$$H_{k+1} = (I - \frac{s_k y_k^T}{y_k^T s_k}) H_k (I - \frac{y_k s_k^T}{y_k^T s_k}) + \frac{s_k s_k^T}{y_k^T s_k}, \quad s_k = x_{k+1} - x_k, \quad y_k = \nabla f_{k+1} - \nabla f_k \quad (8)$$

So the total update for minimax problem is:

Algorithm 1 BFGS for minimization problem

step size α
 $P_{x,k} = -H_{x,k} \nabla_x f(x_k, y_k)$
 $s_{x,k} = \alpha P_{x,k}$
 $x_{k+1} = x_k + s_{x,k}$
 $g_{x,k} = \nabla_x f(x_{k+1}, y_k) - \nabla_x f(x_k, y_k)$
 $H_{x,k+1} = (I - \frac{s_{x,k} g_{x,k}^T}{g_{x,k}^T s_{x,k}}) H_{x,k} (I - \frac{g_{x,k} s_{x,k}^T}{g_{x,k}^T s_{x,k}}) + \frac{s_{x,k} s_{x,k}^T}{g_{x,k}^T s_{x,k}}$

Algorithm 2 BFGS update for maximization problem

step size α
 $P_{y,k} = H_{y,k} \nabla_y f(x_k + 1, y_k)$
 $s_{y,k} = \alpha P_{y,k}$
 $y_{k+1} = y_k + s_{y,k}$
 $g_{y,k} = \nabla_y f(x_{k+1}, y_{k+1}) - \nabla_y f(x_{k+1}, y_k)$
 $H_{y,k+1} = (I - \frac{s_{y,k} g_{y,k}^T}{g_{y,k}^T s_{y,k}}) H_{y,k} (I - \frac{g_{y,k} s_{y,k}^T}{g_{y,k}^T s_{y,k}}) + \frac{s_{y,k} s_{y,k}^T}{g_{y,k}^T s_{y,k}}$

4 Experiment

- In gaussian mean, cqn perform better than gda, worse than cn, but gdqn and qngd perform similar and worse than gda
- In gaussian mean ill, gdqn and cqn perform better than gda but worse than cq, qngd perform worse than gda
- In gaussian covariance, diverge

5 Other Note

- generator is the minimizer which is the slow leader
- discriminator is the maximizer which is the faster follower

- will the update of other side effect the approximation of Hessian
- run the inner problem more times and the Hessian should not be used between iterations
- care about the input of inner problem: - loss fn
- bug: rho when calculating V and H
- take notes